

COMPLETE WATERHALL IoT Capstone Shopping List

To ensure you have **absolutely everything** needed to build, wire, and test a fully functional physical IoT prototype for your final defense, here is the exhaustive, complete list of materials.

Nothing is missing from this list. If you buy these items, you will have a complete working physical rig.

1. Core Electronics & Microcontrollers

The "brains" that will read sensors and send data to your server. * ☐ **ESP32 Development Board (38-pin or 30-pin)** (Qty: 2) - 1 for the main reservoir, 1 for the household meter. * ☐ **Micro-USB or USB-C Cables** (Qty: 2) - Must be *data* cables, not just charging cables, so you can program the ESP32s from your PC.

2. Water Sensors (The Inputs)

The sensors that collect the data your app displays. * ☐ **JSN-SR04T Waterproof Ultrasonic Sensor** (Qty: 1) - Do not buy the standard HC-SR04. The JSN-SR04T is waterproof and won't corrode when placed above the water tank. * ☐ **Analog Turbidity Sensor Module (e.g., TS-300B)** (Qty: 1) - Comes with a probe and a small signal board to measure water clarity. * ☐ **YF-S201 (1/2") Water Flow Sensor** (Qty: 2) - One for a "normal" house, one for a "leaking" house so you can demonstrate the difference during defense.

3. Wiring & Prototyping Supplies

Everything needed to connect the sensors to the microcontrollers safely. * ☐ **830-Tie Point Breadboards** (Qty: 2) - For testing connections before making them permanent. * ☐ **Jumper Wire Kit (120 pcs)** (Qty: 1) - Make sure it includes Male-to-Male, Male-to-Female, and Female-to-Female wires. * ☐ **10k Ohms Resistors** (Qty: 1 small pack) - Needed as pull-up resistors for the water flow sensors. * ☐ **Soldering Iron & Lead-Free Solder Wire** (Qty: 1) - Optional but highly recommended if you want a permanent, stable circuit for your final defense. * ☐ **Waterproof Plastic Enclosure Box** (Qty: 2) - To house the ESP32s and breadboards so water doesn't short circuit your electronics during demonstration.

4. Power Supply

Powering the electronics and the pumps without blowing up the ESP32. * ☐ **5V 2A USB Wall Adapters** (Qty: 2) - Standard phone chargers to power the ESP32s continuously during the defense. * ☐ **12V 2A Power Adapter (DC Barrel Jack)** (Qty: 1) - Specifically to power the water pump (pumps require more power than an ESP32 can provide). * ☐ **DC Female Barrel Jack Adapter** (Qty: 2) - To easily connect the 12V adapter to the pump wires.

5. Plumbing & Physical Rig (The Simulation)

The physical setup you will put on the table during your defense to prove the sensors work with real water. * ☐ **12V Mini Submersible Water Pump** (Qty: 1) - To push water through the pipes and spin the flow sensors. * ☐ **Clear Plastic Storage Box / Aquarium (Approx 10-20 Liters)** (Qty: 1) - This acts as your "Central Reservoir". * ☐ **Small Plastic Pitcher / Bucket** (Qty: 2) - To act as the "households" that catch the water coming out of the pipes. * ☐ **1/2" Clear PVC Tubing (3 Meters)** (Qty: 1) - Clear tubing is best so panelists can see the water flowing. * ☐ **1/2" PVC Ball Valves** (Qty: 2) - Put these before the flow sensors. You can partially close them during defense to simulate "low flow" or a "leak". * ☐ **Teflon Tape** (Qty: 1) - To prevent leaks where the PVC tubing connects to the flow sensors.

6. Testing Tools

- ☐ **Digital Multimeter** (Qty: 1) - Crucial for troubleshooting if a sensor isn't turning on. Helps you verify 5V or 3.3V power is actually flowing.

Pro-Tip for Capstone Defense:

When you set up the physical rig, use **food coloring** in the water (just a tiny drop) or stir up some dirt when

testing the Turbidity sensor to instantly trigger the "Water Safety Alert" on your Worker App dashboard. Panelists love seeing physical changes instantly reflect on the mobile app!